

Notifications Feature in PMS

- ✓ This document includes the Design Details for implementing the Certificate Expiry Notifications feature in PMS Revamp.

UX: [Figma UX](#)

Story: [MOSIP-33095: Notifications \(Partner Admin & all Partners\)](#) CLOSED

In Partner Management Services application, we will add new Spring Batch Jobs which will generate the notifications. Different types of notifications will be generated by different batch jobs.

- Each Job will be triggered as per the Cron Job Schedule set for them in the properties files.
- Each Job will create a new row in the notifications table in DB along with all the notification details.

1. **Root / Intermediate Certificate Expiry Notifications Job**

- To get the list of all root certificates expiring in next 30 days time, invoke the Key Manager endpoint `/getCaCertificates`, with the request body attribute `validTillDate` set to current date + 30 days.
- Next find all partner ids which are active and have the PARTNER_ADMIN role using the keycloak URL `/auth/admin/realms/{realm}/roles/PARTNER_ADMIN/users`
- For each partner who has PARTNER_ADMIN role, check if there is an entry in `pms.partner` table and is it active, if yes, add the notification in the table. The `notification_details_json` will be in below format

```
1  {
2    "certificateId": "",
3    "issuedBy": "",
4    "issuedTo": "",
5    "partnerId": "", //It will be null for Root and Intermediate CA certs
6    "partnerDomain": "",
7    "expiryDateTime": "",
8    "expiryPeriod": "",
9    "certificateType": ""
```

Along with this the `email_id` , `lang_code` of the partner admin into `pms.notifications` table, with type `ROOT_CERT_EXPIRY` or `INTERMEDIATE_CERT_EXPIRY`

- Notification is to be added in the table based on Notification Frequency only. 30 days before the expiry, 15 days before the expiry, 10 days before the expiry, 9 days before the expiry, so on till 0 days before the expiry. So repeat from step a - c based on these intervals. So for any certificate id there may be 13 notifications at most in the table for a specific partner id.
- These same steps can be done for both Root and Intermediate certificates. In the Key Manager endpoint `/getCaCertificates` we can send `caCertificateType` as `root` or `intermediate`
- Once notification is added trigger the email notification. Refer to point #7 below.
- The table will include below columns

Column Name	Data Type	Length	IS_NULL	Comments
<code>id</code>	<code>character varying</code>	128	FALSE	Primary Key
<code>partner_id</code>	<code>character varying</code>	128	FALSE	Partner Id for which this notification is generated
<code>notification_type</code>	<code>character varying</code>	36	FALSE	Allowed Values <code>PARTNER_CERT_EXPIRY</code> <code>ROOT_CERT_EXPIRY</code> <code>INTERMEDIATE_CERT_EXPIRY</code> <code>WEEKLY_SUMMARY</code>

				SBI_EXPIRY API_KEY_EXPIRY FTM_CHIP_CERT_EXPIRY
notification_status	character varying	36	FALSE	Allowed values ACTIVE DISMISSED
notification_details_json	character varying	4000	FALSE	Notification details JSON with below attributes <pre>1 { 2 "certificateDetails": [3 { 4 "certificateId": "", 5 "issuedBy": "", 6 "issuedTo": "", 7 "partnerId": "", 8 "partnerDomain": "", 9 "expiryDateTime": "", 10 "expiryPeriod": "30", 11 "certificateType": "root" 12 } 13], 14 "ftmDetails": null, 15 "apiKeyDetails": null,</pre>

				16 "sbiDetails": null 17 }
email_id	character varying	256	FALSE	The email id given by the partner during registration with PMS.
email_language_code	character varying	36	FALSE	The notification language code selected by the partner during registration with PMS.
email_sent	boolean		TRUE	Flag indicating if Email was sent or not.
email_sent_timestamps	timestamp without time zone		TRUE	The date in ISO format when the notification email was sent.
cr_by	character varying	128	FALSE	
cr_timestamps	timestamp without time zone		FALSE	The date in ISO format when the notification was created.
upd_by	character varying	128	TRUE	
upd_timestamps	timestamp without time zone		TRUE	The date in ISO format when the notification was updated.

2. **Partner Certificate Expiry Notifications Job**

- a. First find all partner ids which have the PARTNER_ADMIN role using the keycloak URL `/auth/admin/realms/{realm}/roles/PARTNER_ADMIN/users`.
- b. Get all the active partners in the table `pms.partner`. Ignore partners which have the Partner Admin role.
- c. Loop through partner id in the batch. Invoke the Key Manager endpoint `/getPartnerCertificate/{partnerCertId}`. Here the `partnerCertId` matches the `certificateAlias` field in the table `pms.partner`. *If the processing is time consuming, we can process the partner ids in a multi threaded manner, so that we can achieve efficiency.*
- d. Check if the expiry date in the certificate if it is after 30 days, after 15 days or within next 10 days.
- e. Add the notification in the table along with all certificate details, email id, `lang_code` of the partner into `pms.notifications` table, with type `PARTNER_CERT_EXPIRY`
- f. Once notification is added trigger the email notification. Refer to point #7 below.

3. **SBI Expiry Notifications Job** - Design will be added in next release

4. **FTM Chip Certificate Expiry Notifications Job** - Design will be added in next release

5. **API Key Expiry Notifications Job** - Design will be added in next release

6. **Weekly Summary Notifications**

- a. In Weekly Summary we have to show number of expiring Partner Certificates, FTM Chip Certificates, SBI, API keys. For this release we will be only including the number of expiring Partner Certificates in the Weekly Summary notification. To create this.
- b. First find all partner ids which have the PARTNER_ADMIN role using the keycloak URL `/auth/admin/realms/{realm}/roles/PARTNER_ADMIN/users`.
Cache the list of Partner Admins.
- c. Get all the active partners in the table `pms.partner`. Ignore partners which have the Partner Admin role.
- d. Loop through partner id in the batch. Invoke the Key Manager endpoint `/getPartnerCertificate/{partnerCertId}`. Here the `partnerCertId` matches the `certificateAlias` field in the table

pms.partner *If the processing is time consuming, we can process the partner ids in a multi threaded manner, so that we can achieve efficiency.*

- e. Above steps a - d are actually same as in **Partner Certificate Expiry Notifications Job**. So add below steps in the same Job.
- f. Check if the expiry date in the certificate if it is within next 7 days.
- g. For each partner who has PARTNER_ADMIN role (cached in step b), check if there is an entry in **pms.partner** table and is it active, if yes, add a notification in the table along with all certificate details, email id, **lang_code** of the partner admin into **pms.certificate_notifications** table, with type **WEEKLY_SUMMARY**.
- h. The **notification_details_json** will be in below format. More details will be saved to support the View Expiring Items page which will be developed in next release

```
1 {
2   "certificateDetails": [
3     {
4       "certificateId": "",
5       "issuedBy": "",
6       "issuedTo": "",
7       "partnerId": "",
8       "partnerDomain": "",
9       "expiryDateTime": "",
10      "expiryPeriod": "7",
11      "certificateType": "partner"
12    },
13    {
14      "certificateId": "",
15      "issuedBy": "",
16      "issuedTo": "",
17      "partnerId": "",
18      "partnerDomain": "",
19      "expiryDateTime": "",
20      "expiryPeriod": "7",
21      "certificateType": "partner"
22    }
23  ],
24  "sbiDetails": [
25    {
26      "sbiId": "",
27      "sbiVersion": "",
28      "partnerId": "",
29      "expiryDateTime": "",
30      "expiryPeriod": "7",
31      "sbiBinaryHash": "",
32      "sbiCreationDate": ""
33    }
34  ],
35  "ftmDetails": [
36    {
```

```

37         "certificateId": "",
38         "issuedBy": "",
39         "issuedTo": "",
40         "partnerId": "",
41         "partnerDomain": "",
42         "expiryDateTime": "",
43         "expiryPeriod": "7",
44         "certificateType": "ftm",
45         "ftmId": "",
46         "make": "",
47         "model": ""
48     },
49 ],
50     "apiKeyDetails": [
51     {
52         "apiKeyName": "",
53         "partnerId": "",
54         "expiryDateTime": "",
55         "expiryPeriod": "7",
56         "policyGroup": "",
57         "policyName": ""
58     }
59 ]
60 }

```

Once notification is added trigger the email notification. Refer to point #7 below.

7. Send Email Notification

- This will be a method in a utility class which will be triggered every time after notification is added in DB.
- Get the notification details from the `pms.notifications` table based on `notificationId`.
- For each notification check if the email was sent or not. If no, then trigger an email notification.
- For email notification fetch the email template from the Master Data endpoint `/v1/masterdata/templates/{langCode}/{templateTypeCode}`. Lang code and email id is available in the `pms.certificate_notifications` table. The template type code should be taken from the properties file. Example:
`email.notification.root.cert.expiry.template =`
`ROOT_CERT_EXPIRY_TEMPLATE`
- Once template is fetched, populate the dynamic values in it like the Certificate Id, Issued To etc.
- Next use the `kernel notifications` service to trigger the email.

- After email notification is triggered, update the col `email_sent` as true and also update the `email_sent_dtimes`.
- Proper logs should be added to track how many email notifications were sent. Also failures and exceptions should be logged.

8. Delete Past Notifications Job

1. This job will check all the notifications that were created before a configurable number of days, say 60.
2. All these notifications will be hard deleted from the DB.
3. Proper logs should be included.
4. Also add `data-archive` scripts to handle deleting the past notifications.
5. Also make this job configurable so that it can be disabled in the env, where `data-archive` scripts will take care of this.

API Endpoints:

1. New GET endpoint `/v1/partnermanager/notifications` will be added.
 - This endpoint will get all the notifications from the `pms.notifications` table.
 - The notifications fetched will be for all the partner ids which are linked to the logged in user id, which is sent as a Header in the Authorization token.
 - This endpoint will be used to get the notifications to be displayed in the Notification Panel as well as in the View All Notifications page.
 - This will include optional request parameters to support pagination and filtering.
 - The request parameters will include optional paramter, `notificationType`. Allowed values for the same will be `root`, `intermediate`, `partner`, `weekly`, `sbi`, `ftm-chip`, `apikey`. If the `notificationType` is `root`, `intermediate` or `weekly` then the `PARTNER_ADMIN` role is expected in the token. If it is not available then error will be thrown.
 - Check details in [Stoplight Documentation](#)
2. New PATCH endpoint `/v1/partnermanager/notifications/{notificationId}` will be added.
 - This endpoint will be used to handle the Dismiss action by the user.
 - Check details in [Stoplight Documentation](#)

3. Add a new optional request parameter `expiryPeriod` to the GET endpoint `/v1/partnermanager/trust-chain-certificates`. The expiry period will act as a filter and its value will be in days. The endpoint should return all trust certificates which are expiring during the given expiry period.
 - This endpoint will be used to populate the Count of expiring certs in the corresponding Dashboard Cards.
 - This endpoint will directly get the certificate details from Key Manager and not from the `pms.notifications` table.
 - Check details in [Stoplight Documentation](#)
4. Add a new optional request parameter `expiryPeriod` to the GET endpoint `/v1/partnermanager/partners/partner-certificates-details`. The expiry period will act as a filter and its value will be in days. The endpoint should return all partner certificates (for the logged in user) which are expiring during the given expiry period.
 - This endpoint will be used to populate the Count of expiring certs in the corresponding Dashboard Card.
 - This endpoint will directly get the certificate details from Key Manager and not from the `pms.notifications` table.
 - Check details in [Stoplight Documentation](#)
5. New GET and PUT `/v1/partnermanager/users/notifications-seen-timestamp` will be added.
 - Add new column `notifications_seen_dtimes` in the `pms.user_details` table.
 - New GET endpoint `/v1/partnermanager/users/{userId}/notifications-seen-timestamp` which will get the status and timestamp of when the notifications were last viewed by the user in the PMS portal. This will be used to show the red alert along with notifications icon in the Header whenever a new notification is generated.
 - New PUT endpoint `/v1/partnermanager/users/{userId}notifications-seen-timestamp` which will update the status and timestamp of when the notifications were last viewed by the user in the PMS portal.
 - Check details in [Stoplight Documentation](#) and [here](#)

